

Technology Stack

Date	27 June 2026
Team ID	[Enter Team ID]
Project Name	HDI Predictor
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Flask	Provides a responsive and user-friendly interface for entering country indicator details and displaying prediction results.
2	Backend / Server-Side	Python, Flask, Scikit-learn	Handles business logic, processes user inputs, loads the trained machine learning model, and generates real-time country development index predictions.
3	Database / Data Storage	Flat CSV Files	Stores country indicator information and historical data efficiently with lightweight filesystem storage.
4	Cloud / Hosting / Deployment	Docker, Flask	Deploys the trained machine learning model in a containerized environment to enable secure and portable web application hosting.
5	Version Control & CI/CD	Git, GitHub	Maintains version control, tracks changes in the ML training pipeline and web server codebase, and simplifies project versioning.
6	Third-Party APIs / Other Tools	Pandas, NumPy, Matplotlib, Pickle	Supports data preprocessing, data visualization, and machine learning model serialization.